

Final Project Write-Up

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Econ. 575 - Jonathan B. Hill

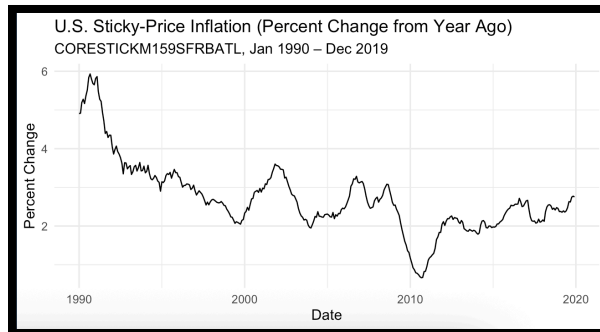
UNC – Chapel Hill

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PROPOSAL

This project forecasts U.S. sticky-price inflation excluding food and energy (seasonally adjusted). The goal is to predict future monthly values of sticky-price inflation using its past values. The series measures the percent change from one year ago in consumer prices that adjust slowly, excluding food and energy. The data are monthly observations from the Federal Reserve Economic Data archive (FRED).

The sample spans January 1990–December 2019 ($T = 360$). Because the series is already expressed as a percent change from a year ago, it is naturally seasonally adjusted; however, the series will still be detrended, as there is a long-run trend component. The dependent variable (y) is the monthly sticky-price inflation rate. There are no additional explanatory variables at this stage, so the model will be univariate. The series fluctuates around a stable mean with moderate persistence and no clear long-term trend, suggesting that an ARMA model is appropriate for forecasting.



I will model the monthly sticky-price inflation rate y_t with a univariate ARMA(p, q):

$$y_t = u + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t, \quad \text{with } \epsilon_t \sim (0, \sigma^2)$$

I will estimate models for several combinations of p and q , ranging from 0 to 4, and select the preferred p^* and q^* subject to white noise errors based on the lowest AIC. The residuals will be checked to ensure they resemble white noise using the Ljung–Box Q-test. If two models have similar AIC values, I will select the simpler model with fewer parameters.

To evaluate forecast performance, I will conduct a rolling-window analysis in which the in-sample window equals $T-100$, generating 100 one-step-ahead forecasts for each model. The model with the lowest mean squared forecast error will be chosen as the final forecasting model. Using this model, I will produce 1-, 3-, 6-, and 12-month-ahead point forecasts and 95% forecast intervals. After

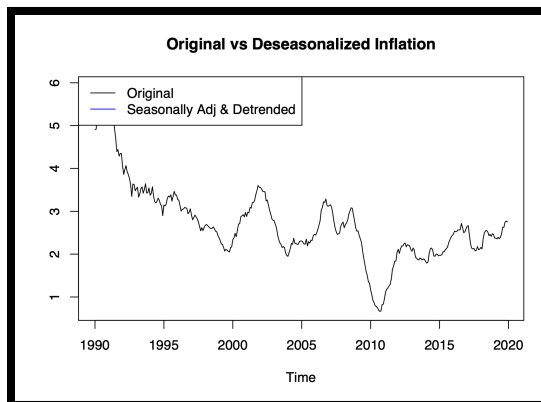
estimating the preferred ARMA model, I will test the residuals for autocorrelation and conditional heteroskedasticity. If the residuals exhibit heteroskedasticity, I will estimate a GARCH(1,1) model to capture volatility and generate volatility forecasts as an extension.

The main hypothesis is that sticky-price inflation is persistent, implying that at least one autoregressive coefficient ϕ is positive and statistically significant ($H_0 : \phi_i = 0$ vs. $H_A : \phi_i \neq 0$). I expect a low-order ARMA model, such as ARMA(1,1) or AR(2), to provide accurate short-horizon forecasts and to outperform other simple baseline models in terms of mean squared forecast error

PRELIMINARY RESULTS

(i) Seasonal Adjustment and Detrending

The sticky-price inflation series (Jan 1990–Dec 2019) was seasonally adjusted and linearly detrended. The estimated seasonal component was effectively zero, indicating that the data from FRED are already seasonally adjusted. The adjusted series was therefore identical to the original. Figure 1 shows the overlapping lines. The ADF test on the adjusted series produced a p-value of 0.01, suggesting that the series is stationary. The deseasonalized and detrended series will be used for ARMA modeling.



ii) Selecting Optimal ARMA

Table 1 reports AIC and Ljung–Box Q-test p-values for each (p,q) combination. The optimal model was selected by minimizing AIC while requiring residuals to be white (Q-test $p > 0.20$). The chosen model is **ARMA(2, 1)** with AIC = **-651.73** and Q-test $p = 0.73$, satisfying the white noise condition.

p	q	AIC	Q test p-val
0	0	285.02	0.000
1	0	-625.62	0.000
2	1	-651.73	0.730

3	1	-651.38	0.729
1	4	-634.51	0.583
4	4	-678.36	0.016

(iii) Full-Sample Estimation

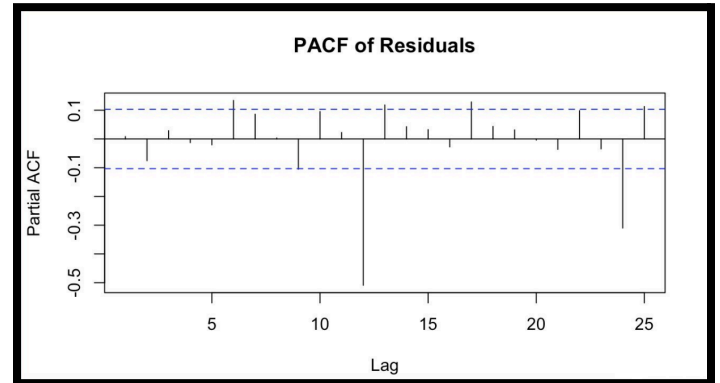
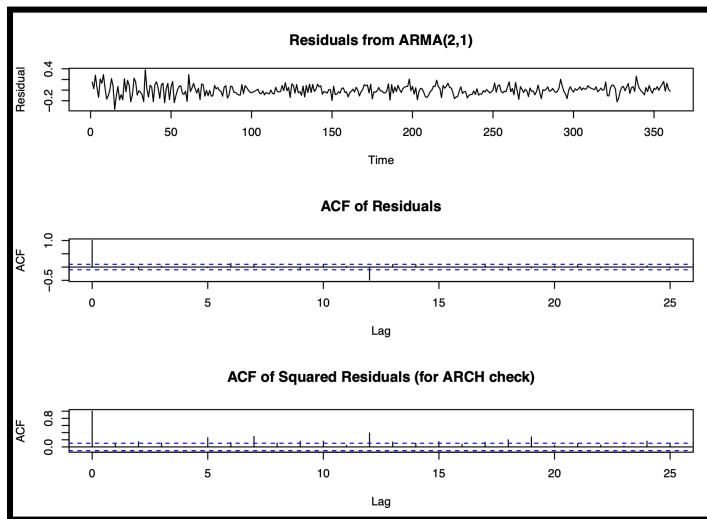
The ARMA(2,1) model was estimated using [farmax.roll](#) on the full sample (January 1990–December 2019). The AR(1), AR(2), and MA(1) coefficients are all highly significant ($p < 0.001$), indicating strong autoregressive persistence and mean reversion dynamics. The intercept is not statistically significant. The model achieves an R^2 of 0.9795 and an AIC of -651.73 . The Ljung–Box Q-test with $H = k + 5$ ($H = 8$) yields $p < 2.2e-16$, suggesting residual autocorrelation remains. Despite this, the ARMA(2,1) specification provides the lowest AIC and is the best-fitting model. It will be used for the forecasting steps that follow.

parameter	estimate	SE	t-val	p-val
constant	0.02913	0.16830	0.17	0.8642
AR(1)	1.90640	0.03920	48.63	0.0000
AR(2)	-0.91370	0.03880	-23.55	0.0000
MA(1)	-0.75790	0.05830	-13.00	0.0000
R^2	0.9795			
AIC	-651.73			
Q-test p-val	$< 2.2e-16$			

(iv) Residual Diagnostic Tests

Residuals from the ARMA(2,1) model were analyzed for serial correlation and conditional heteroskedasticity. The PACF and ACF from the residual plots show no significant autocorrelation, and both behave like white noise around the mean. There are no visible patterns, so the mean dynamics were captured well. The Ljung–Box Q-test ($H = 8$) yielded a p -value $< 2.2e-16$, indicating significant residual autocorrelation, showing evidence of ARCH effects. The ARCH test with 12 lags produced a Chi-squared statistic of 92.27 ($p = 1.79e-14$), suggesting strong evidence of time-varying volatility. These results imply that, while the ARMA(2,1) captures the main dynamic pattern in inflation, a GARCH-type extension may be necessary to properly model conditional heteroskedasticity.

v) Residual Plot



FORECASTS AND FINAL RESULTS

(i) MSFE Evaluation

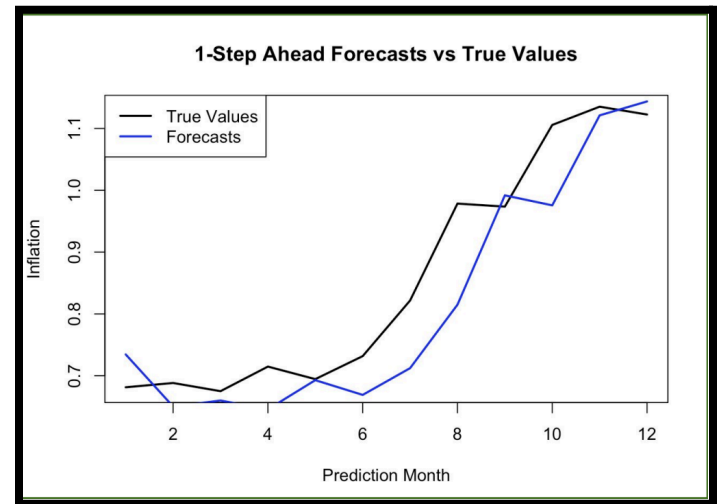
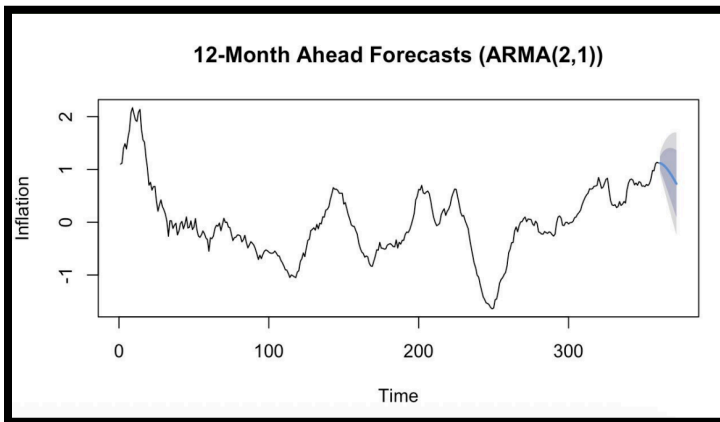
Using a 120-month rolling window, I computed 1-step-ahead MSFEs for several ARMA models. The ARMA(2,1) model produces the **lowest MSFE (0.0079)**, outperforming higher-order alternatives such as ARMA(3,1) and ARMA(2,2), as well as simpler structures like ARMA(1,1) and AR(2). Although some of the larger models have slightly lower in-sample AIC values, they do not generalize as well out-of-sample and tend to overfit. In addition, the negative AR(2) term (≈ -0.94) pulls the overall persistence back down. The sum of coefficients is approximately 1, creating a higher MSFE value.

Model	MSFE
ARMA(2,1)	0.0079
ARMA(3,1)	0.0082
ARMA(1,1)	0.0084
ARMA(2,2)	0.0080
AR(2)	0.0090

(ii) ARMA(2,1) Forecasting

After running a multi-step 12-month ahead forecast, it presented relatively stable forecasts and mean-reverting behavior. The widening of the confidence intervals represents increased uncertainty, the

further into the future you forecast. These results are consistent with stationary inflation dynamics. In addition, I ran a 1-step ahead recursive forecast over the past 12 months. The blue line shows the model's 1-step ahead predictions, and the black line shows the true inflation values. The forecasts slightly underpredict inflation, especially in months with large movements. This is consistent with the earlier ARCH LM test, which indicated strong volatility clustering, further emphasizing the need for a GARCH-type model.



(iii) GARCH(1,1) Model for volatility

Given the strong ARCH effects detected in the ARMA(2,1) residuals (LM-test p-value $< 1e-14$), I estimated a GARCH(1,1) model on the seasonally adjusted and detrended series. The estimated model is:

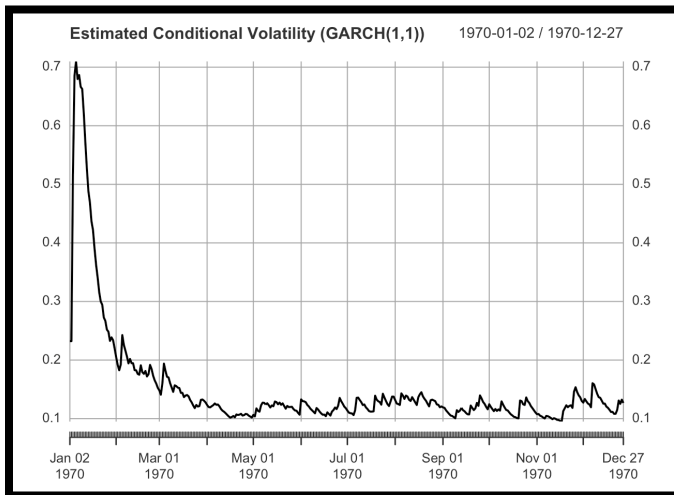
$$y_t = \mu + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \theta_1 \epsilon_{t-1}$$

Parameter	Estimate
$\omega = 0.001197$	baseline variance
$\alpha_1 = 0.08046$	short-run shock effect (ARCH term)
$\beta_1 = 0.84436$	persistence of volatility (GARCH term)
$\alpha_1 + \beta_1 = 0.9248$	high volatility persistence

Since $\alpha + \beta \approx 0.92$, volatility is **highly persistent** but < 1 , so the model is stable. This confirms sticky-price inflation volatility tends to cluster but **eventually mean-reverts**. Mean equation coefficients (AR and MA) remain close to the ARMA(2,1) estimates.

iv) Estimated Conditional Volatility Plot

Early sample periods exhibit **higher volatility**, then gradually stabilize over time. The decline in volatility fits macroeconomic patterns: sticky-price inflation fluctuates less aggressively in the 2000s compared to the early 1990s. The model captures these time-varying variance dynamics, unlike ARMA-only models.



v) 12-Step Ahead Forecasts CI's

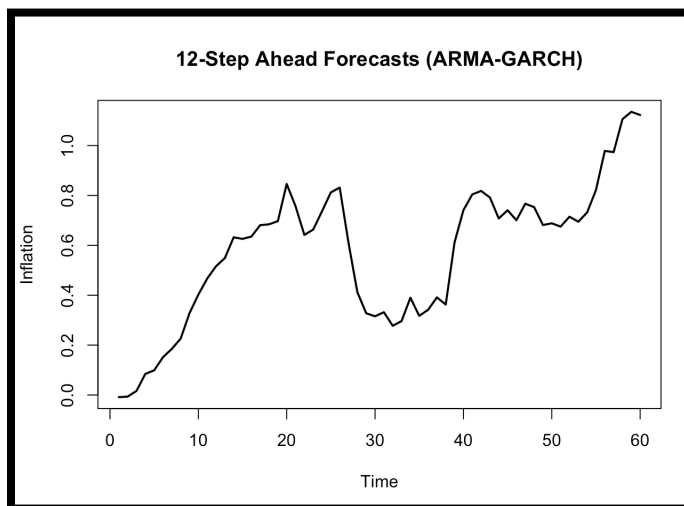
The 95% confidence intervals widen steadily as the forecast horizon increases from 1 to 12 months. This widening is expected: uncertainty accumulates the further ahead I forecast, especially in a series that exhibits volatility clustering, as identified by the ARCH LM test. Overall, the intervals capture the degree of uncertainty in forecasting sticky-price inflation and visually demonstrate the influence of volatility dynamics.

Month Ahead	Forecast	Interval
1	1.659676	(1.4211268, 1.898226)
2	1.642845	(1.4036258, 1.882064)
3	1.616250	(1.3764136, 1.856087)
4	1.580625	(1.3402180, 1.821031)
5	1.536717	(1.2957843, 1.777649)

6	1.427101	(1.2438693, 1.726704)
7	1.427101	(1.1852357, 1.668967)
8	1.362926	(1.1206469, 1.605205)
9	1.293522	(1.0508617, 1.536183)
10	1.219643	(0.9766298, 1.462656)
11	1.142026	(0.8986873, 1.385365)
12	1.061392	(0.8177524, 1.305031)

vi) 12-Step Ahead Forecasts Using ARMA–GARCH

The black line shows the **true historical sticky-price inflation** for the most recent 60 months. The ARMA-GARCH model **predicts the next 12 months**, accounting for the autoregressive structure *and* time-varying volatility, with confidence intervals widening over longer horizons due to volatility persistence. The model predicts **a gradual decline in inflation over the next year** (from about 1.66 down to 1.06). The point forecasts are smooth and slowly trending downward, emphasizing a slow-moving, persistent series. GARCH improves the variance modeling



Including a GARCH(1,1) structure substantially improves the modeling of inflation volatility, producing more accurate confidence intervals while maintaining the strong mean dynamics captured by ARMA(2,1).